# The wedge for Lecture 4

- Demand and supply curves are the results of buyers' and sellers' decision making. That is, to pursue comparative advantages, they make transaction and thus make decision on quantity demanded and supplied considering the presence of opportunity costs.

- Elasticity of demand and supply can then be defined and discussed according to the shapes of these demand and supply curves.

- The main question is how those buyers and sellers make decisions about what and how many quantities they buy or produce that results in the demand and supply curves we discuss in previous lectures.

- Let's start from how those buyers (consumers) make decisions.

# Lecture 4: 消費者選擇與需求曲線的導出

**(Consumer Choices and Derivation of Demand Curves)**

# Outline

- 4-1
  - 4-1.1: Utility and Preference
  - 4-1.2: Assumption about Consumers’ Preference
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- 4-3
  - 4-3.1: Derivations of Demand Curves
  - 4-3.2: Substitution and Income Effects
- 4-4: The Analyses of Optimal Consumers’ Choices under Two Special Consumers’ Preference

## 4-1.1: 效用與偏好

**(Utility and Preference)**

- 效用：耗用資源後，消費者滿足的程度
  **(Utility: The measure of happiness or satisfaction after consuming resources, like money, food,…etc.)**

- 偏好：有限資源下，在耗用資源以滿足慾望時，消費者取捨的基準
  **(Preference: When the resources are limited, preference refers to the consumers optimal choices between alternatives. That is, it describes what the consumers want)**

- **The results of the decision on quantity demanded:**
  - When resources are limited, a rational decision on quantity demanded obeys the “law of demand” in general. That is, for the product X and other things being equal given a certain time interval, its quantity demanded and its price move reversely.

- **The process of making decisions on quantity demanded:**
  - The demand decision is the one made by consumers to maximize their own utility with the presence of budget constraints
  - Budget constraints describe what the consumers can afford
  - Utility describes the consumers’ preference of what they want.

- **Utility Function**
  - **It describes the relation between consumption bundle and preference**

- Assume that a buyer consumes n types of products and it can be denoted by $(Q_1^D, Q_2^D, \ldots, Q_n^D)$, where $Q_i^D$ represents the quantity of the $ith$ product consuming by the buyer.

- After consuming the n types of products, the consumer's satisfaction can be measured to the level U. The U can be described as

$$\mathrm{U} = h(Q_1^D, Q_2^D, \ldots, Q_n^D),$$

where $h$ is the utility function.

- The function $h$ describes the way how the consumption bundle $(Q_1^D, Q_2^D, \ldots, Q_n^D)$ maps to the corresponding satisfaction level U.

It is a kind of description about a consumer's preference.

- **The Definition of Marginal Utility, MU**

  - Assume the original consumption bundle is $(Q_1^D, Q_2^D)$. The buyer then consume the product 1 more to make his consumption bundle $(Q_1^D + \Delta Q_1^D, Q_2^D)$

  - His original consumption can reach the satisfaction level $U_b = h(Q_1^D, Q_2^D)$;

    his new consumption portfolio can reach the satisfaction level $U_a = h(Q_1^D + \Delta Q_1^D, Q_2^D)$

  - To capture how the increment in consumption of the product 1, $\Delta Q_1^D$, affect the buyer's utility, we use marginal utility

$$\frac{U_a - U_b}{\Delta Q_1^D} = \frac{\Delta U}{\Delta Q_1^D} = MU_1$$

    $MU_1$ is called the marginal utility of product 1

  - In the same sense, $MU_2$ is the marginal utility of product 2.

- For example:

  if a consumer's utility function is $\mathrm{U} = h(Q_1^D, Q_2^D)$, where $Q_1^D$ is the quantity of consumption for fruits and $Q_2^D$ is for meat.

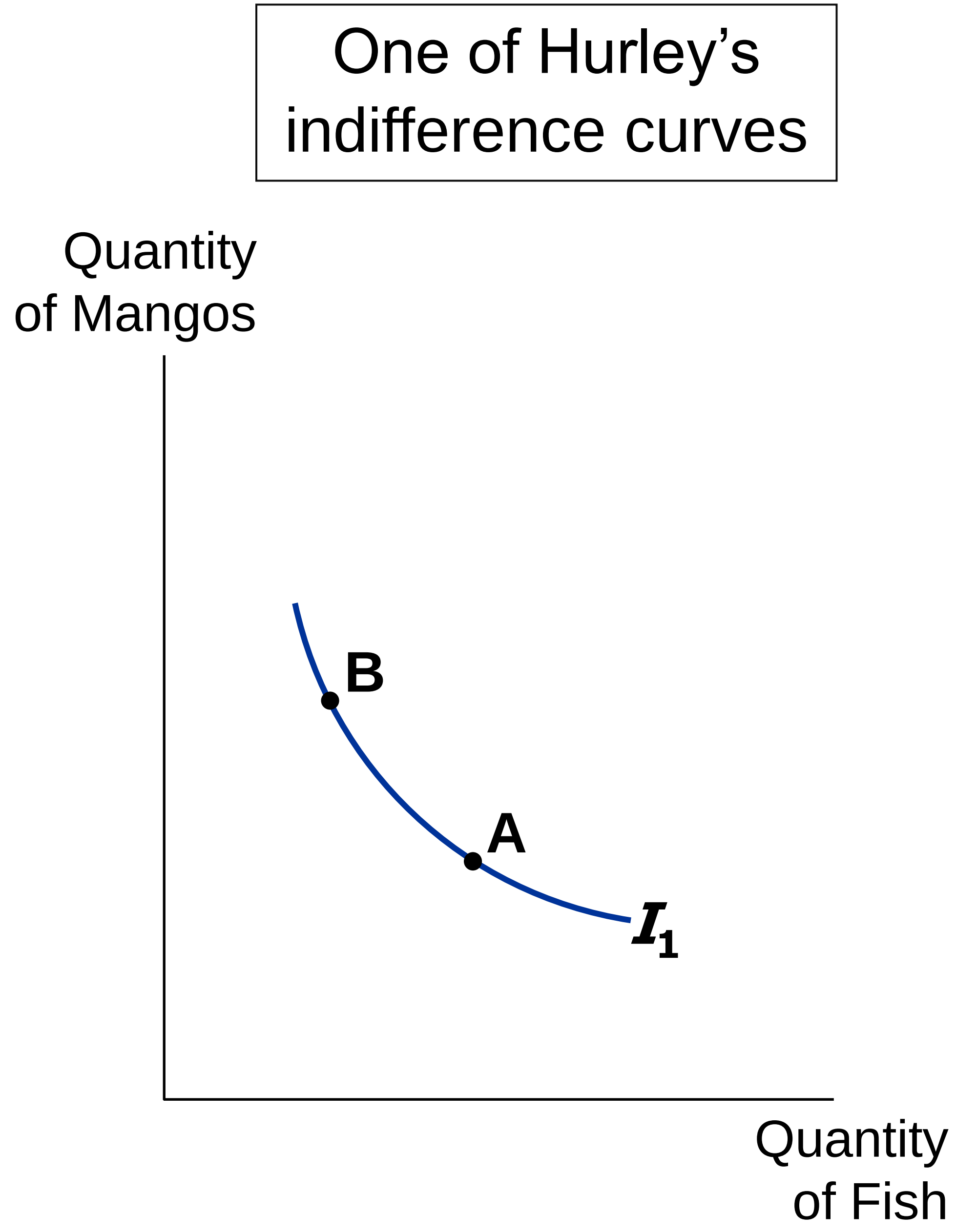
  If he consumes 1kg of fruits and 0.5 kg of meats, then his satisfaction level is $U_b = h(1{,}0.5) = 100$

  If he increases his consumption for meats to 0.6kg, then his satisfaction level is $U_a = h(1{,}0.6) = 102.$ Thus, the marginal utility of meat is

$$MU_2 = \frac{U_a - U_b}{\Delta Q_2^D} = \frac{102-100}{0.6-0.5} = 20$$

- **Indifference curve**: shows consumption bundles that give the consumer the same level of satisfaction

- **A**, **B**, and all other bundles on $I_1$ make Hurley equally happy: he is *indifferent* between them.

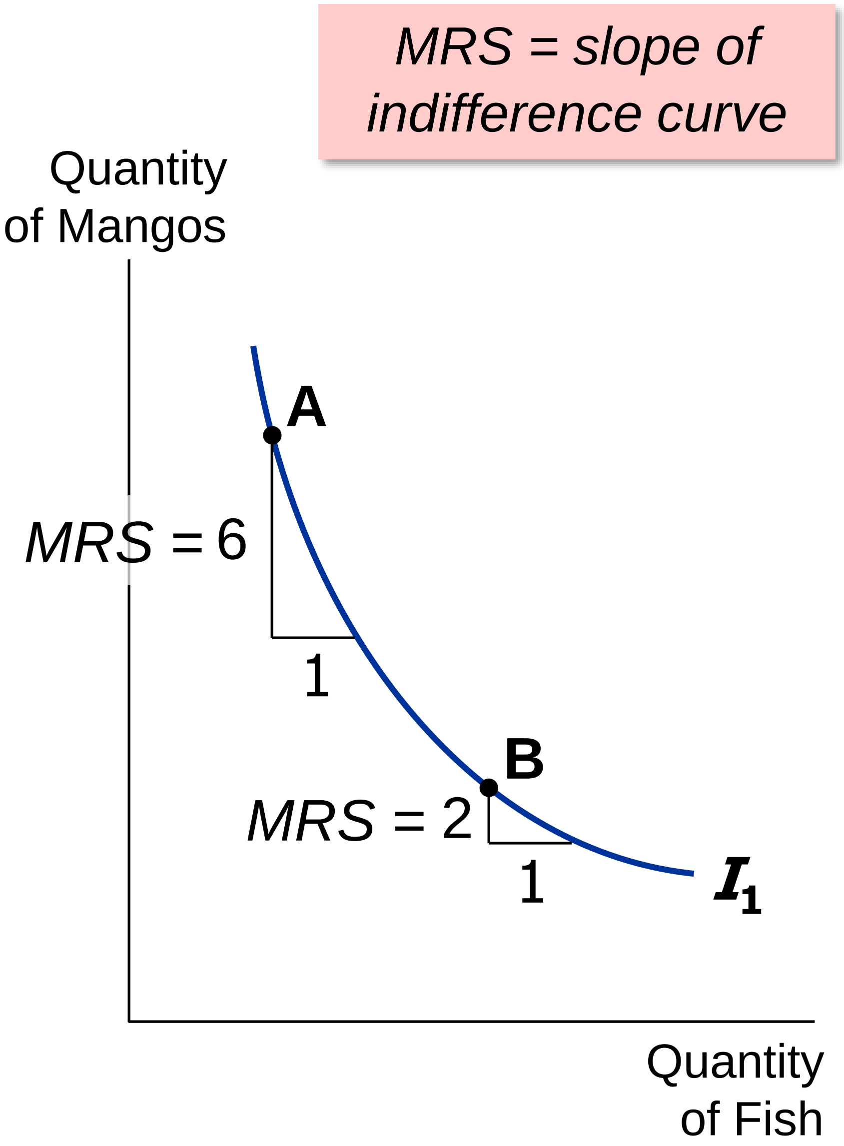


- **Marginal rate of substitution (MRS)**:

  the rate at which a consumer is willing to trade one good for another.

Hurley's MRS is the amount of mangos he would substitute for another fish.

MRS falls as you move down along an indifference curve.



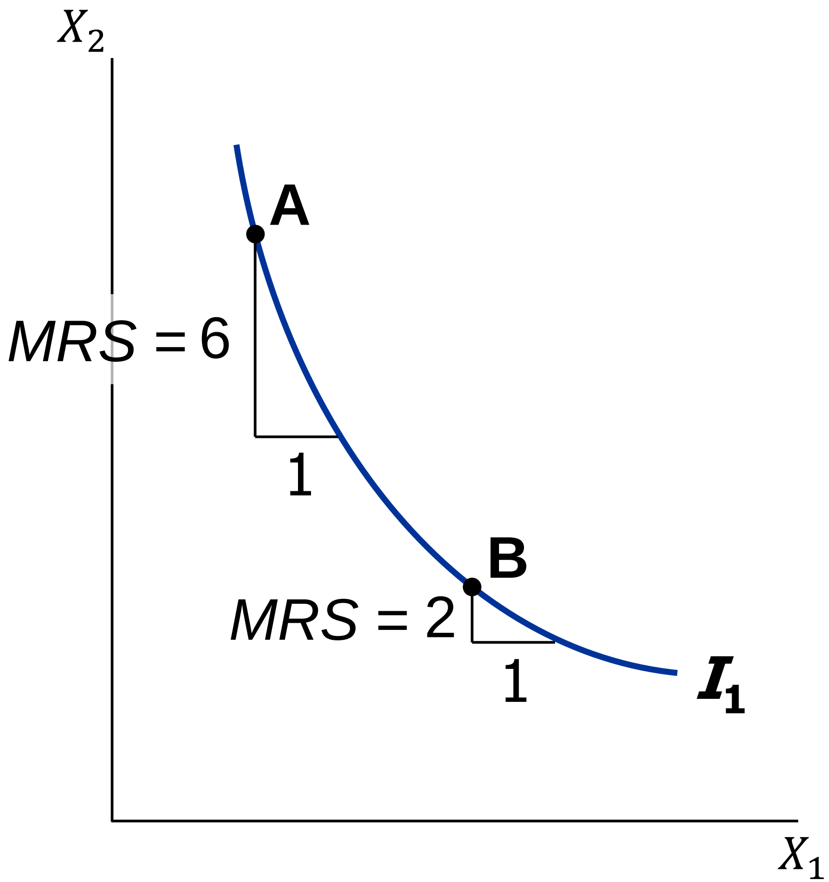
- Notice especially that the MRS at point A, $\mathrm{MRS_A}$, is approximately

$$\mathrm{MRS_A} = \left|\frac{-6}{1}\right| = \left|\frac{\Delta X_2}{\Delta X_1}\right|$$

$$= \left|\frac{\Delta U/\Delta X_1}{\Delta U/\Delta X_2}\right|$$

$$= \left|\frac{MU_1}{MU_2}\right|$$

*MRS = slope of indifference curve*



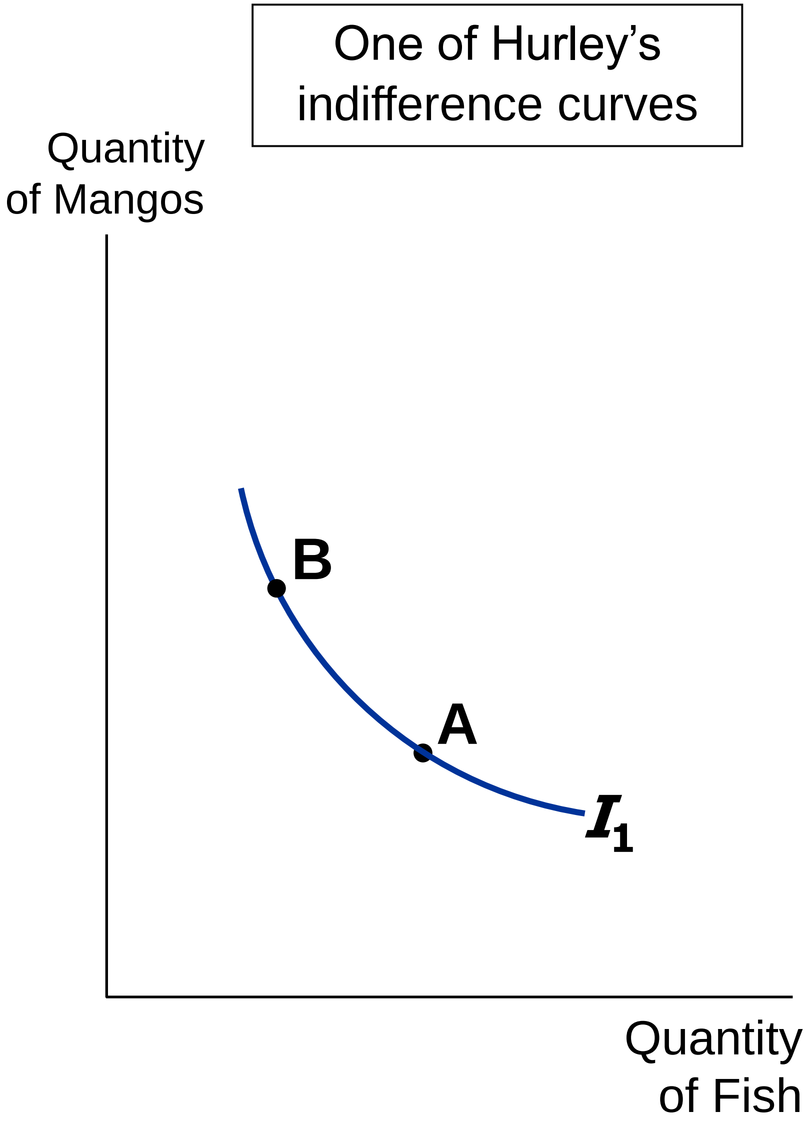
# 4-1.2: 對消費者偏好常做的假設

**(There are four properties of indifference curves)**

# 1. Indifference curves are continuous and smooth.

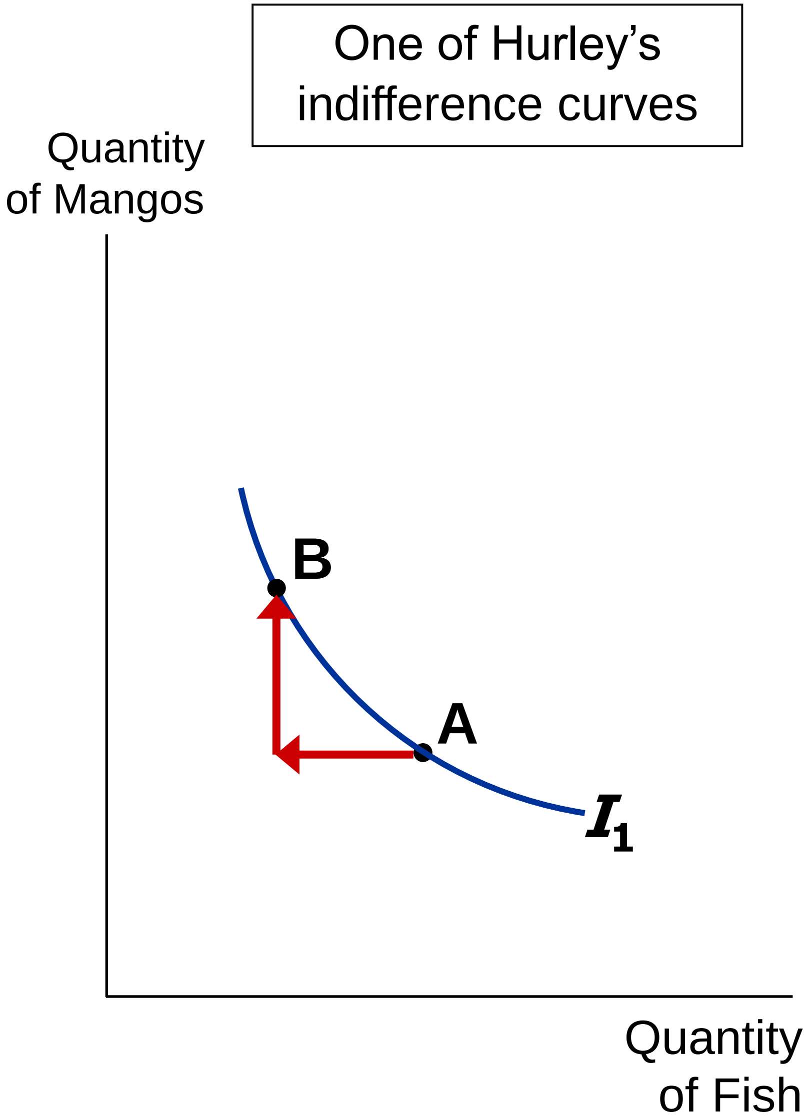
- It means that any kind of products can be divided into tiny pieces, and the consumer can identify the difference between the utility if consuming more pieces and that if consuming fewer pieces .

- It thus can define marginal utility.



# 2-1. Indifference curves are downward-sloping.

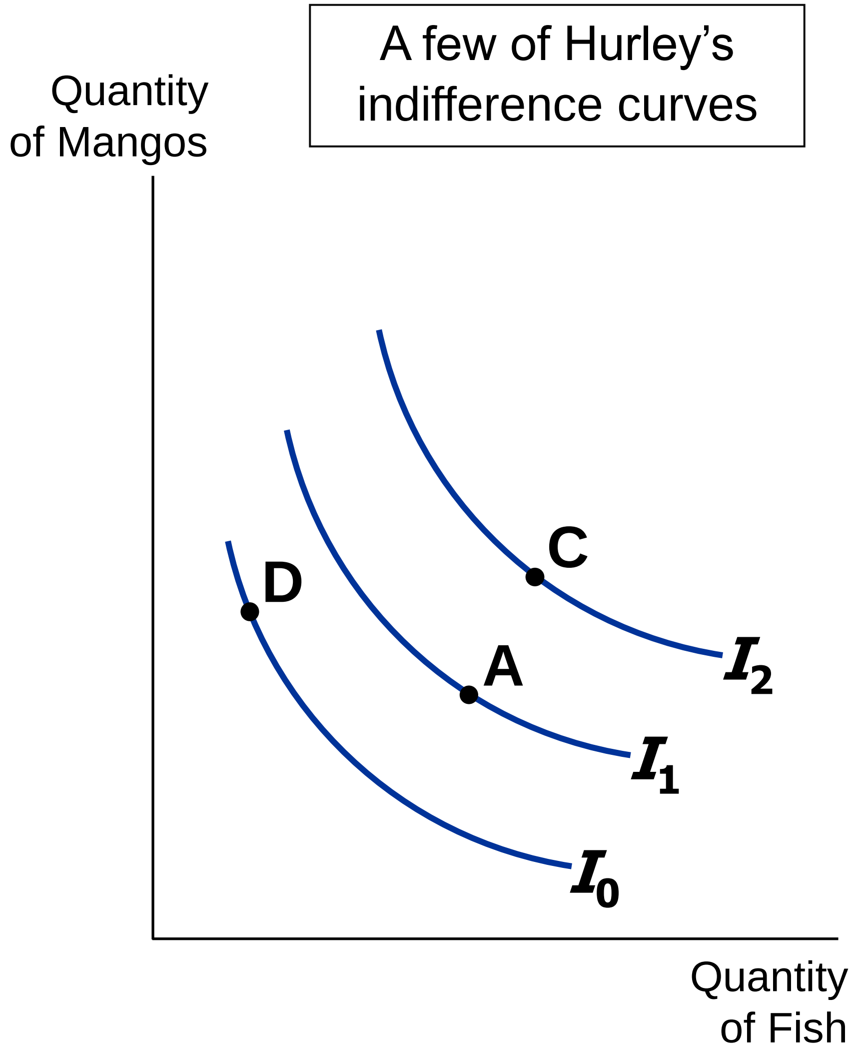
- If the quantity of fish is reduced, the quantity of mangos must be increased to keep Hurley equally happy.



# 2-2. Higher indifference curves are preferred to lower ones.

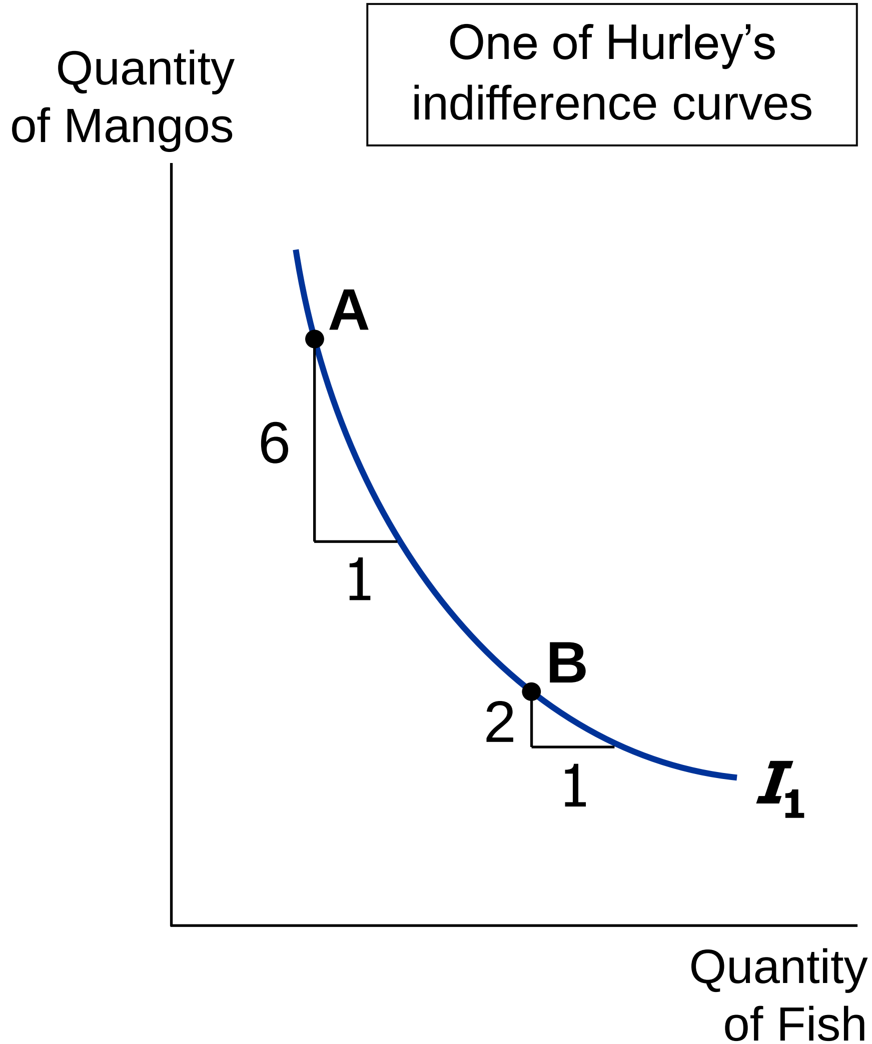
- Hurley prefers every bundle on $I_2$ (like **C**) to every bundle on $I_1$ (like **A**).

- He prefers every bundle on $I_1$ (like **A**) to every bundle on $I_0$ (like **D**).



# 3. Indifference curves are bowed inward.

- Hurley is willing to give up more mangos for a fish if he has few fish (**A**) than if he has many (**B**).



# 4. Indifference curves cannot cross.

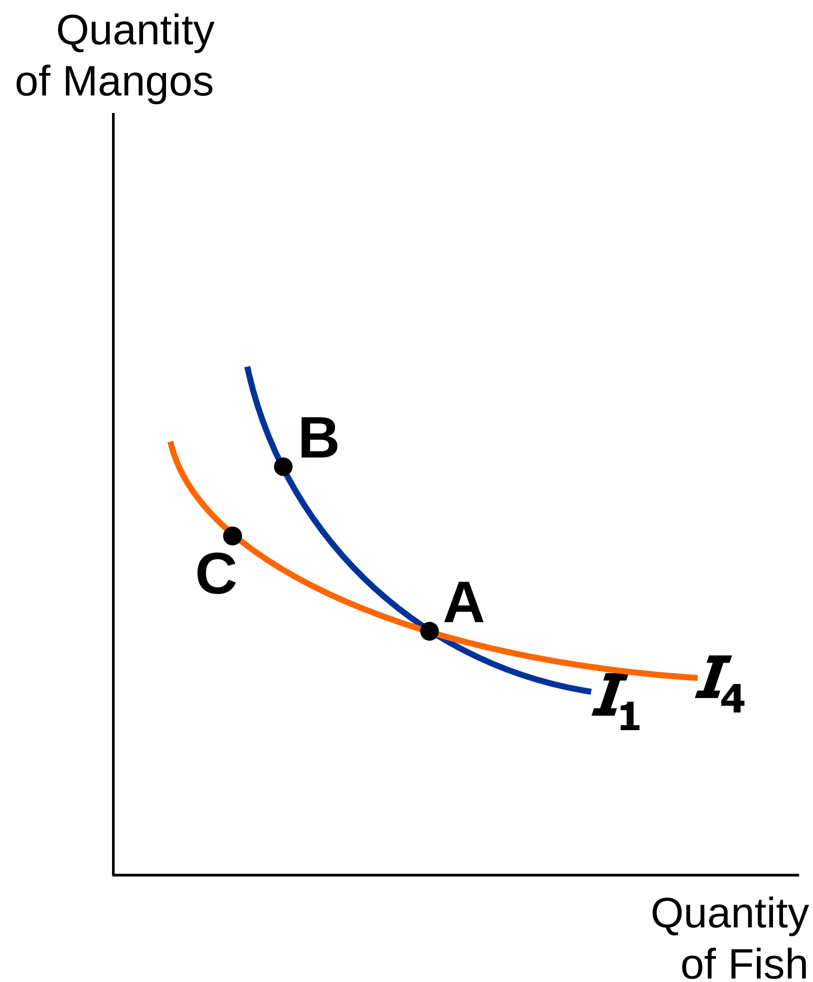
- Suppose they did.
- Hurley should prefer **B** to **C**, since **B** has more of both goods.

- Yet, Hurley is indifferent between **B** and **C**:

– He likes **C** as much as **A** (both are on $\boldsymbol{I_4}$).

– He likes **A** as much as **B** (both are on $\boldsymbol{I_1}$).

Hurley's indifference curves



# Brief Summary

- Generally speaking, an utility function describes the relation between a consumption bundle and a consumer's preference.

Step 1: Given a consumption bundle, an utility function can measure the the consumer's satisfaction level once the he consumes the bundle.

Step 2: If the utility function can measure the satisfaction level for every consumption bundle, then we have indifference curve if the same satisfaction level is connected

Step 3: In addition to satisfaction level, a consumer's preference can also be described by
(a) marginal utility: it can be defined according to the utility function
(b) marginal rate of substitution: it can be defined according to the indifference curves

- A rational consumer

(1) has continuous and smooth indifference curves

(2) has downward-sloping indifference curves; higher indifference curves are preferred to the lower ones

(3) has indifference curves that are bowed inward

(4) has indifference curves not being able to cross

# In-class exercise 4-1

- 令$X_1$表包子消費個數，$X_2$表香蕉消費根數。某人最喜歡的消費組合為$(Q_1^D, Q_2^D) = (3,2)$，吃的太多或太少效用均不及消費(3,2)時的水準。請畫出該消費者可能的無異曲線形狀

  (Suppose that $X_1$ and $X_2$ indicate the consumption volume of bun and banana and a consumer consumes only buns and bananas. The consumer's optimal consumption bundle is $(Q_1^D, Q_2^D) = (3,2)$. This is the only bundle that can maximize his utility. More or less consumption of buns and bananas cannot reach such utility level. Please draw the consumer's indifference curve.)

- 若消費者對$(Q_1^D, Q_2^D)$的偏好如上所述，請問其無異曲線是否符合「負斜率」、「凸向原點」的特性?

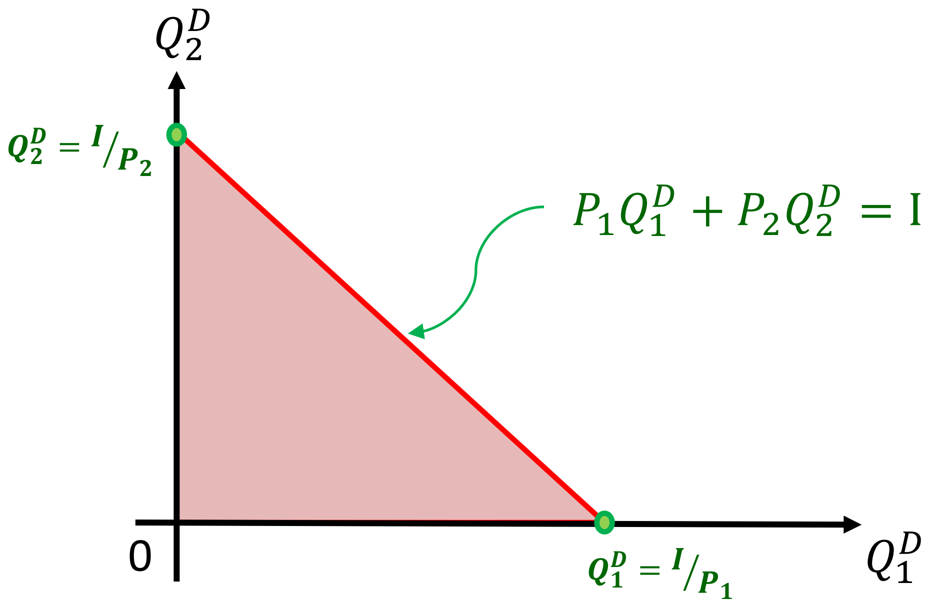
  (Would the indifference curves have negative slope and be bowed inward?)

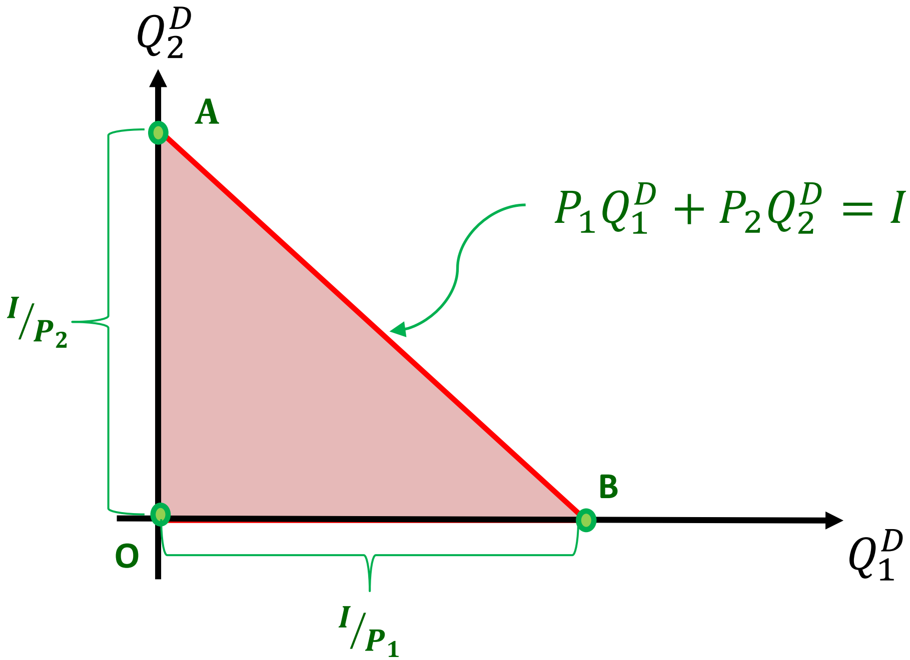
E_Lecture4_W1_Q2

# 4-2: A Consumer's Choice Considering His Preference and Budget Constraint

- An utility function and the corresponding indifference curves describe a consumer's preference.
- One of the most important constraints the consumer faces as he make choices is his budget.
- To analyze how the consumer makes choice, we need to take his preference and budget constraints into consideration.

- **Describing budget constraint mathematically:**
  - Assume that a consumption bundle contains two kinds of products with price per unit $P_1, P_2$
  - If a consumer decides to buy $Q_1^D$ and $Q_2^D$ units, then his total payout is $P_1 Q_1^D + P_2 Q_2^D$
  - Suppose that the consumer has total income $\mathrm{I}$
  - The budget constraint is $P_1 Q_1^D + P_2 Q_2^D \leq \mathrm{I}$






- The absolute value of the slope for $\overline{AB}$ is

$$\frac{I/_{P_2}}{I/_{P_1}} = \frac{P_1}{P_2}$$

Given that the consumer spends all of his income buying the two goods, the decrement in buying one unit of the first good leads to increment in buying $^{P_1}/_{P_2}$ unit of the second good.

It is the market prices of the two products that guide the consumer the way how to face tradeoff.

- The region of ▲ABO is the consumption bundle that the consumer can afford

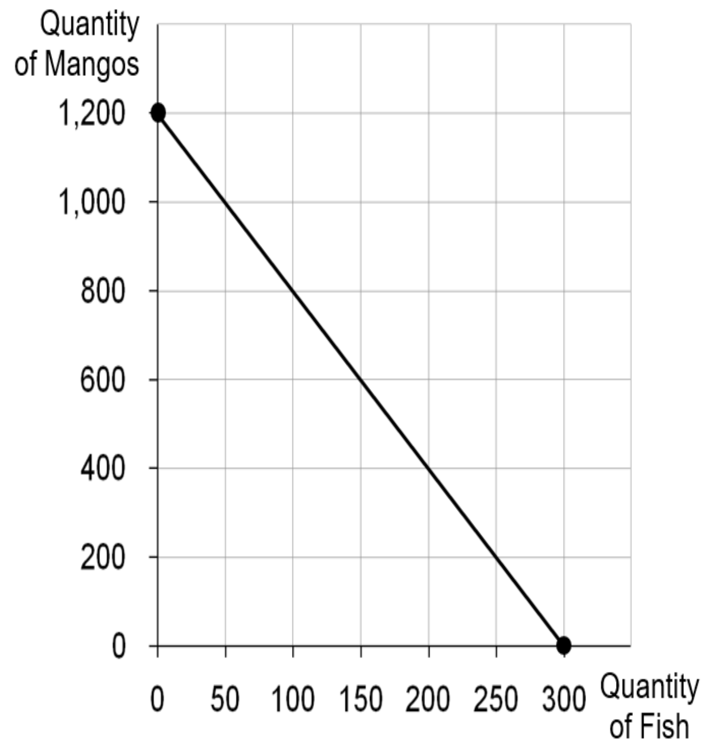
# Another easier case illustrating how the consumer chooses considering his budget constraint

Assume that:

Hurley's income: $1200

Prices: $P_F$ = $4 per fish, $P_M$ = $1 per mango.

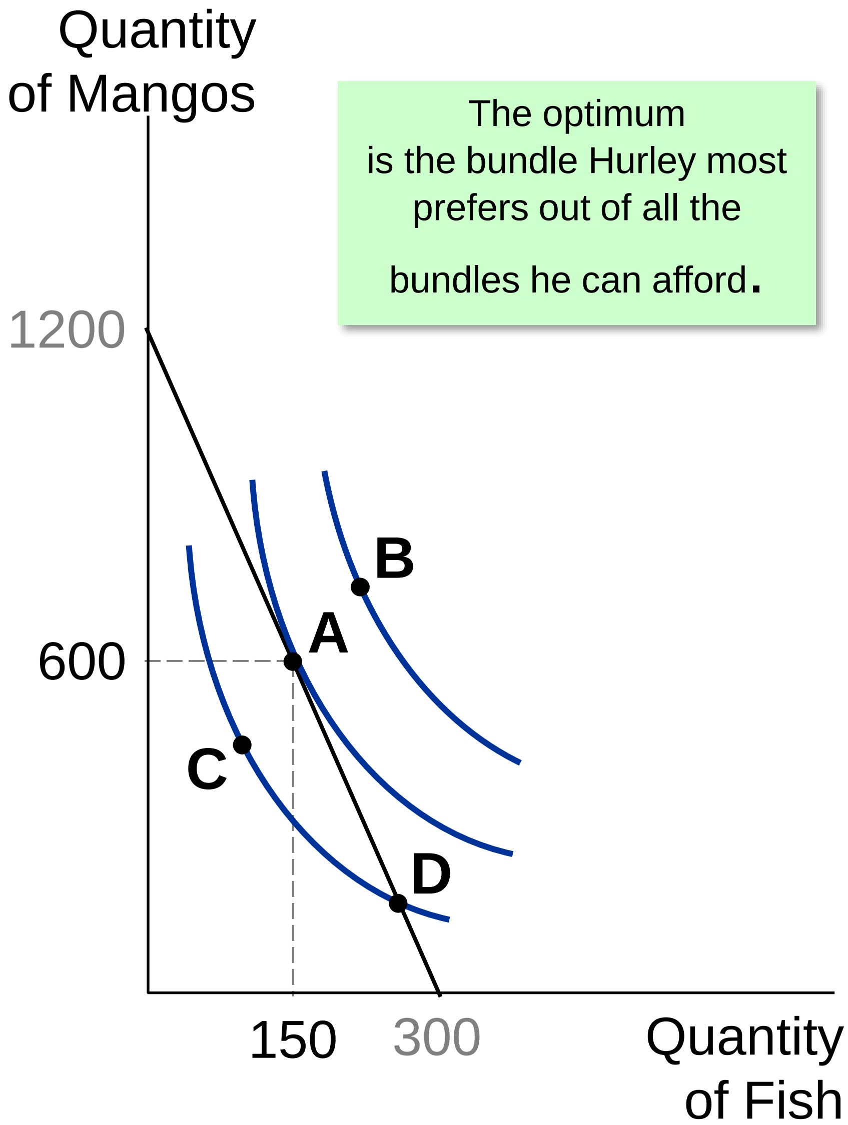
The budget constraint will be



- **A is the *optimum*:**
  the point on the budget constraint that touches the highest possible indifference curve.

- Hurley prefers **B** to **A**, but he cannot afford **B**.

- Hurley can afford **C** and **D**, but **A** is on a higher indifference curve.



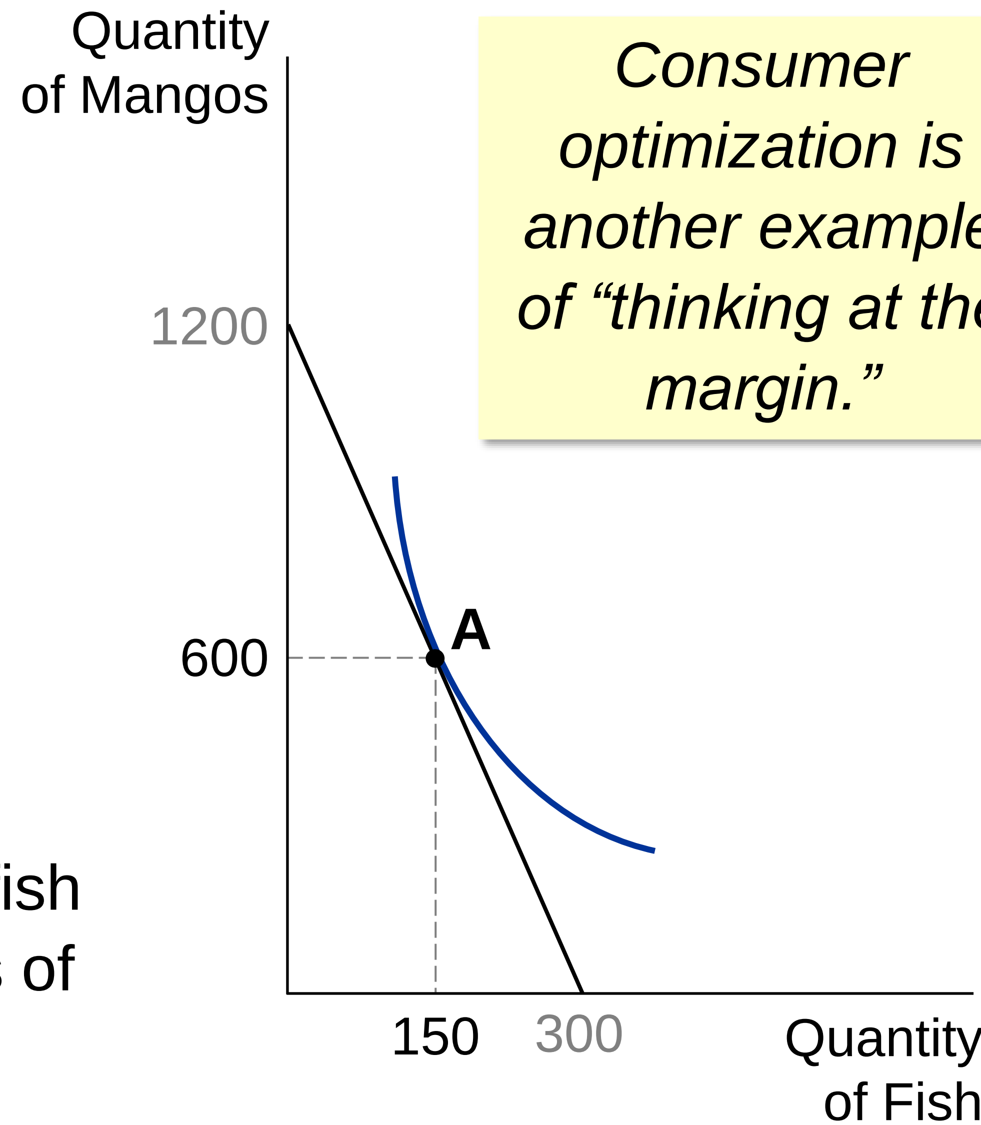

The optimum is the bundle Hurley most prefers out of all the bundles he can afford.

- At the optimum, slope of the indifference curve equals slope of the budget constraint:

$$\text{MRS} = \boldsymbol{P_F} / \boldsymbol{P_M}$$

marginal value of fish (in terms of mangos)

price of fish (in terms of mangos)




*Consumer optimization is another example of "thinking at the margin."*

- And we also know that

$$\text{MRS} = \left|\frac{\text{MU}_\text{F}}{\text{MU}_\text{M}}\right|.$$

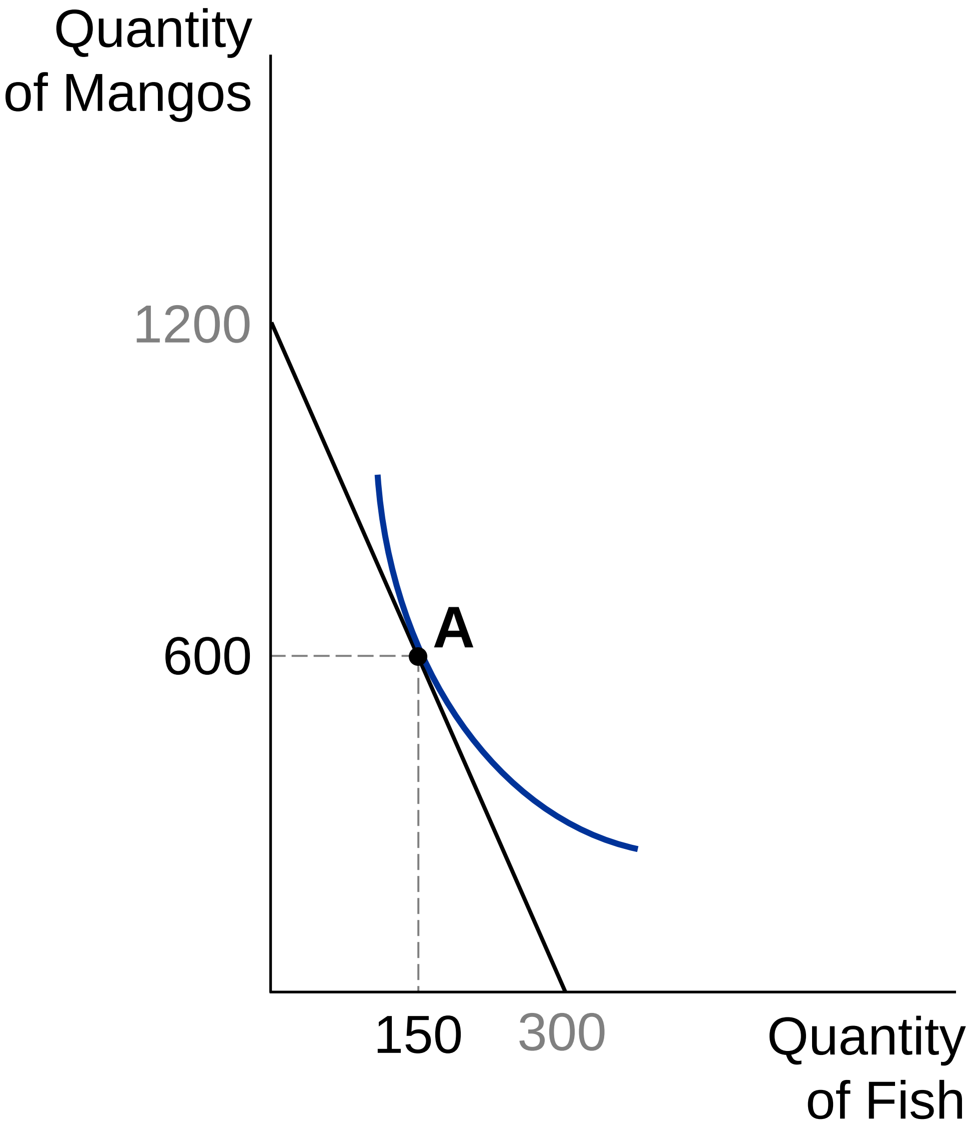
Thus, at the optimum

$$\text{MRS} = \left|\frac{\text{MU}_\text{F}}{\text{MU}_\text{M}}\right| = \frac{P_F}{P_M},$$

leading to

$$\frac{|\text{MU}_\text{M}|}{P_M} = \frac{|\text{MU}_\text{F}|}{P_F}.$$

That implies the marginal utility per dollar spent on mangos equals to marginal utility per dollar spent on fish



# Brief Summary

- The price of the products and consumers' income determines the budget constraints and affordable consumption bundles
- Given that the consumption bundles are affordable, the optimal choice is the bundle being able to reach the highest satisfaction level. That is, the indifference curve that touches the budget constraint.
- The optimal consumption choice here in this lecture has some properties as follows

  (1) Given the types of products, optimal choice is the choice of the volume of the selected products

  (2) The optimal consumption bundles will be on the budget constraint

  (3) MRS is equal to the slope of the budget constraint (the relative price)

  (4) In terms of the optimal consumption bundle, the marginal utility per dollar spent on one good in the bundle equals to marginal utility per dollar spent on the other good

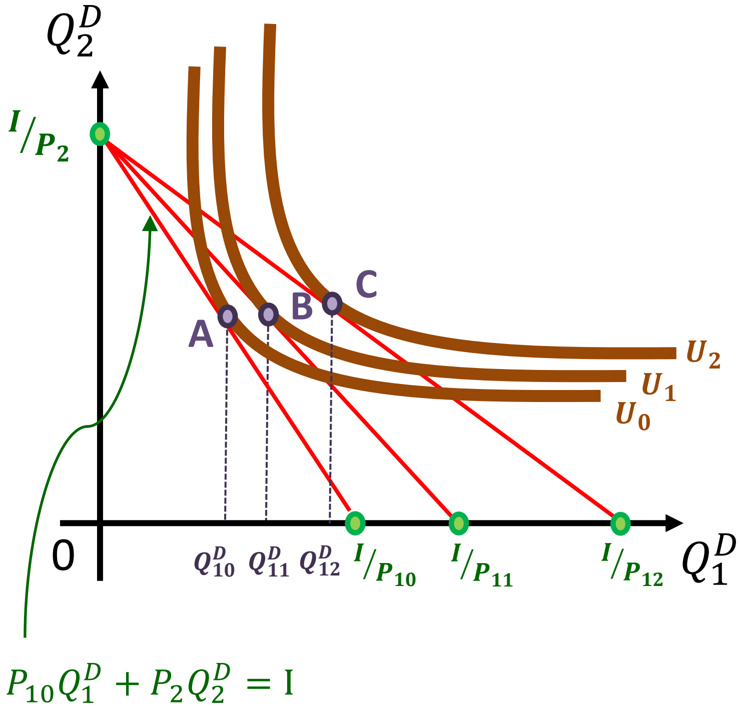
# 4-3.1: Derivations of Demand Curves

- Demand Curves for Product X：
  Other things being equal, e.g., consumers' income, the demand curve illustrates the relation between price $P_X$ and quantity demanded $Q_X^D$.

- Assume a consumer's consumption bundle contains only two products and we take his preference, his income $\mathrm{I}$, and the price of the second product $P_2$ as given. In the following slides, we will discuss the consumer's optimal choice of his consumption bundle $(Q_1^D, Q_2^D)$ under different prices of the first product $P_1$ . That will describe the relation between $P_1$ and $Q_1^D$ from the consumer's point of view.

- As $P_1$ changes, the corresponding optimal choice of consumption bundle $(Q_1^D, Q_2^D)$ also changes. If we connect all of the $P_1$ and $Q_1^D$, it will be the demand curve for the consumer.

- **The procedure of how a consumer determines his quantity demanded considering price of the product and his budget constraint :**



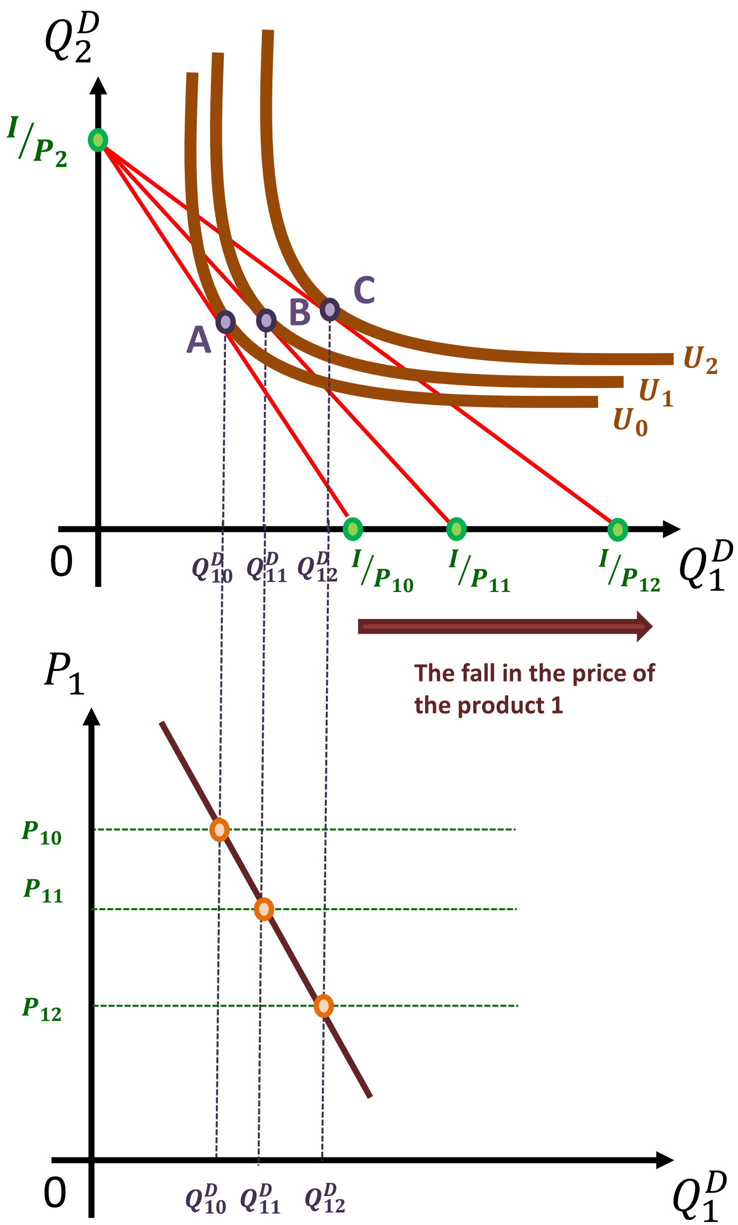

- Given that the price of the product 2 is $P_2$ and a consumer's income is I ; when price of the product 1 is $P_{10}$, his budget constraint is $P_{10}Q_1^D + P_2Q_2^D = \mathrm{I}$.

- When the consumer's utility reaches $U_0$, his indifference curve is tangent to the budget constraint at point A. At this optimum, the price $P_{10}$ corresponds to an optimal quantity demanded $Q_{10}^D$ from the consumer's point of view.

- When the price of the product 1 decreases from $P_{10}$ to $P_{11}$, the region describing the affordable consumption bundles broadens if other things being equal. $U_1$, the higher level of utility will be reached and the new indifference curve is tangent to the new budget constraint at point B. At this optimum, the price $P_{11}$ corresponds to an optimal quantity demanded $Q_{11}^D$

- If the price decreases further to $P_{12}$, the same discussion leads to an optimal quantity demanded $Q_{12}^D$

- **Optimal decision on quantity demanded and the results of the decision:**




- Given that the price of the product 2 is $P_2$ and a consumer's income is I, the left figure shows how a consumer make decision on his quantity demanded

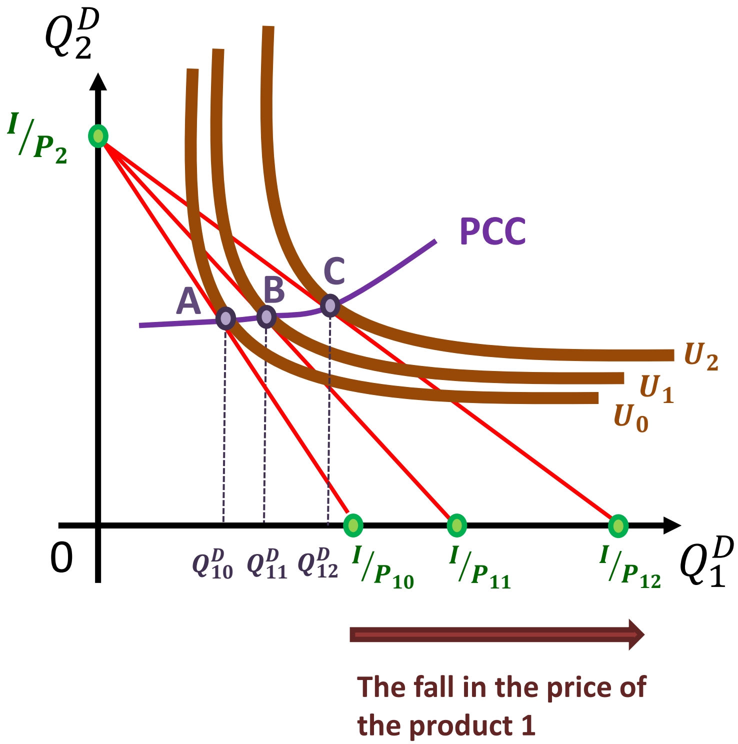
- It can be observed that the optimal decision on quantity demanded will push the utility level up as the price of the product 1 decreases. Note that $U_0 \prec U_1 \prec U_2$

- Shift our focus on the coordinate of price vs. quantity demanded

- Map the prices and the optimal decision on quantities demanded for product 1 from the upper coordinate to the lower

- Connect all of the prices and the quantities demanded resulting from optimal decisions. We then can obtain the curve capturing how the consumer's optimal decision changes as the price of the product 1 changes

- It is the demand curve for the consumer.




- The way how the changes in prices affect the consumer's decisions can be observed through the trajectory comprised by A, B, C.
- The trajectory is called Price-Consumption Curve, PCC. It captures how the consumer's decisions are influenced by price changes given other things being equal.

# 4-3.2: 替代效果與所得效果

(The Substitution and Income Effects)

- 當其他條件不變，當商品1的價格改變所帶來的消費者最適消費決策的改變，當中有替代效果與所得效果

  (Given other things being equal, the changes in a consumer’s optimal decisions on his quantity demanded resulting from price changes contains two effects, the substitution and income effects.)

# Another Interpretation for the Income and Substitution Effects

- Recall that this is Hurley's decision.

Her income I = $1200,

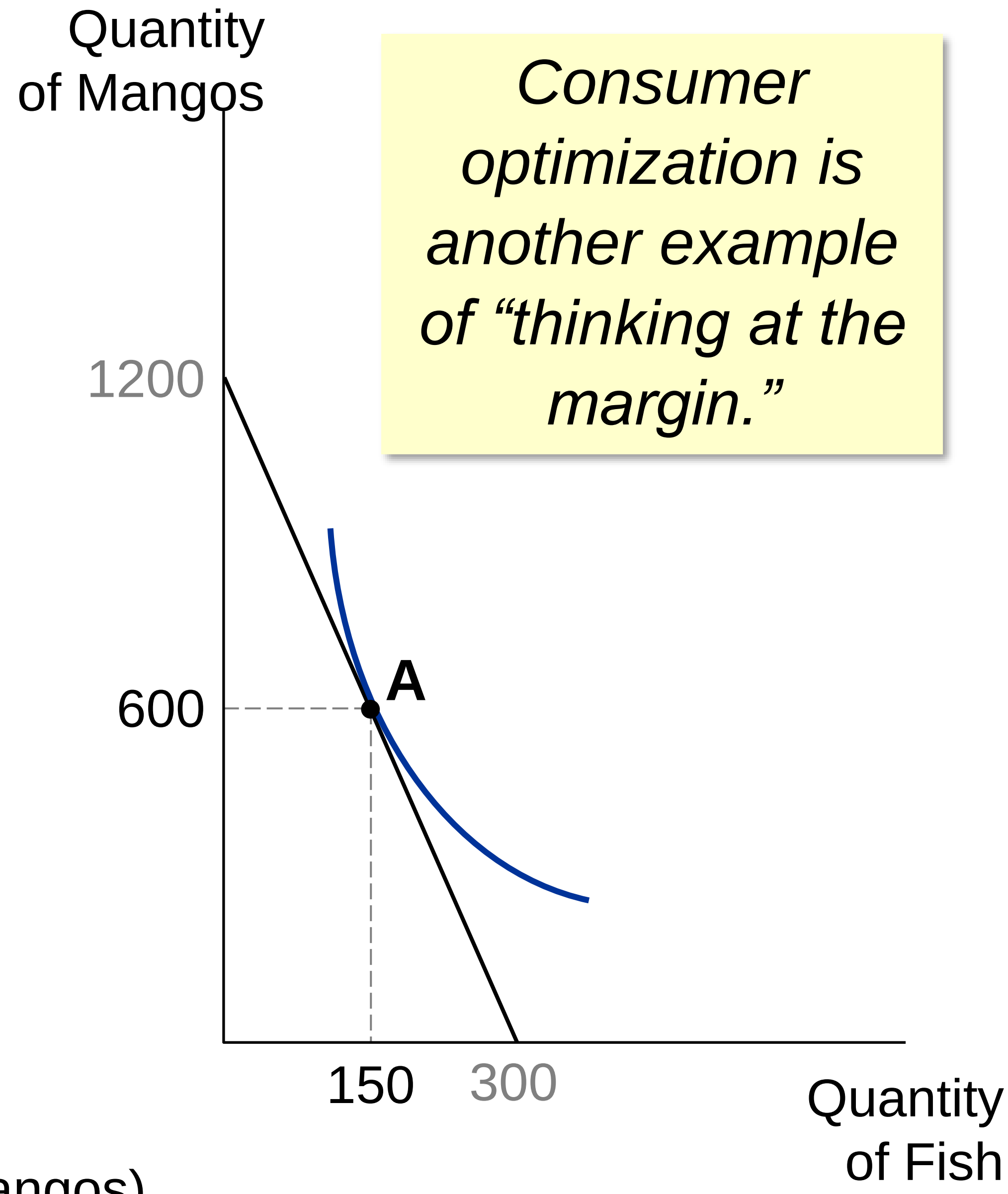
$$P_F = \$4, P_M = \$1$$

At the optimum, slope of the indifference curve equals slope of the budget constraint:

$$\text{MRS} = \boldsymbol{P_F / P_M}$$

marginal value of fish (in terms of mangos)

price of fish (in terms of mangos)




*Consumer optimization is another example of "thinking at the margin."*

A fall in the price of fish has two effects on Hurley's optimal consumption of both goods

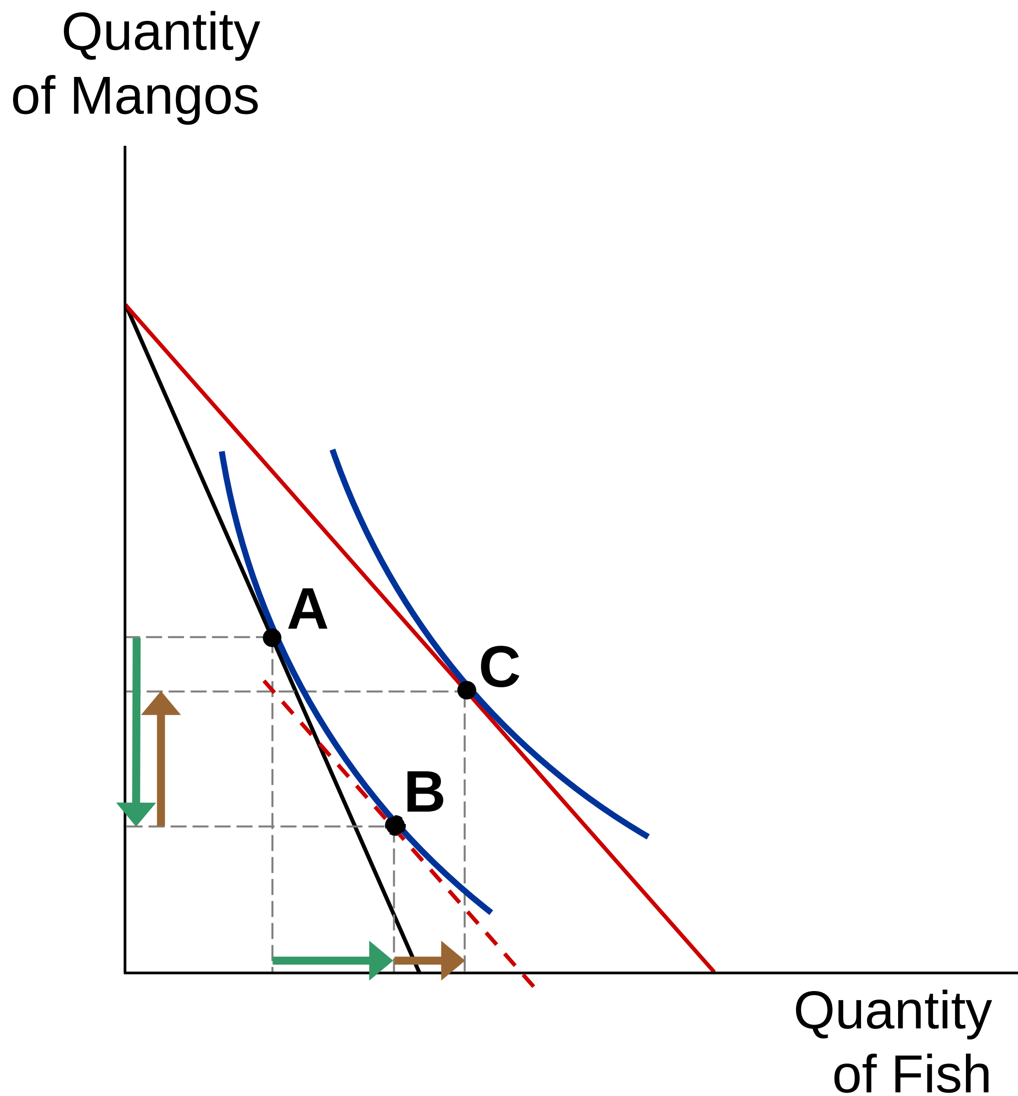
- Income effect
  - A fall in $P_F$ boosts the purchasing power of Hurley's income: buy more mangos and **more fish** (utility level will be higher)
- Substitution effect
  - A fall in $P_F$ makes mangos more expensive relative to fish: Hurley buys fewer mangos and **more fish** (given utility level)
  - Economists called this the negative substitution effect (effect that buy the cheaper good more)

- Initial optimum at **A**.

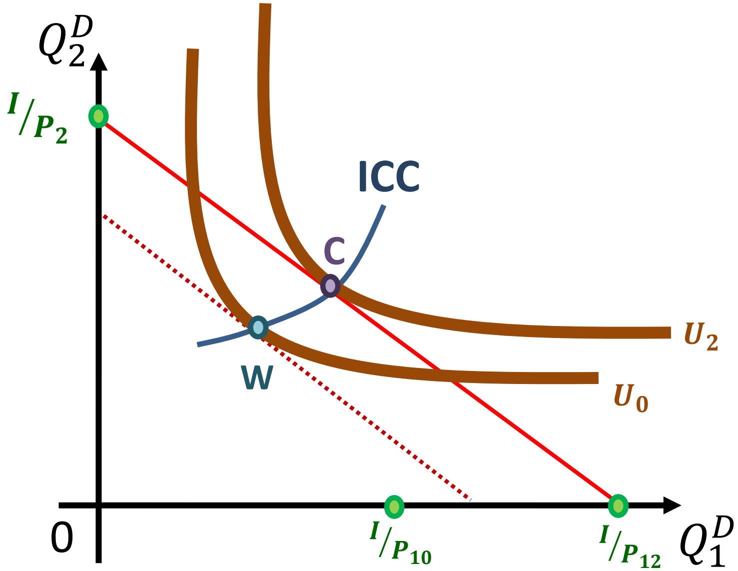
- $\boldsymbol{P_F}$ falls.

***Substitution effect:***
from **A** to **B**,
buy more fish and fewer mangos.

***Income effect:***
from **B** to **C**,
buy more of both goods.



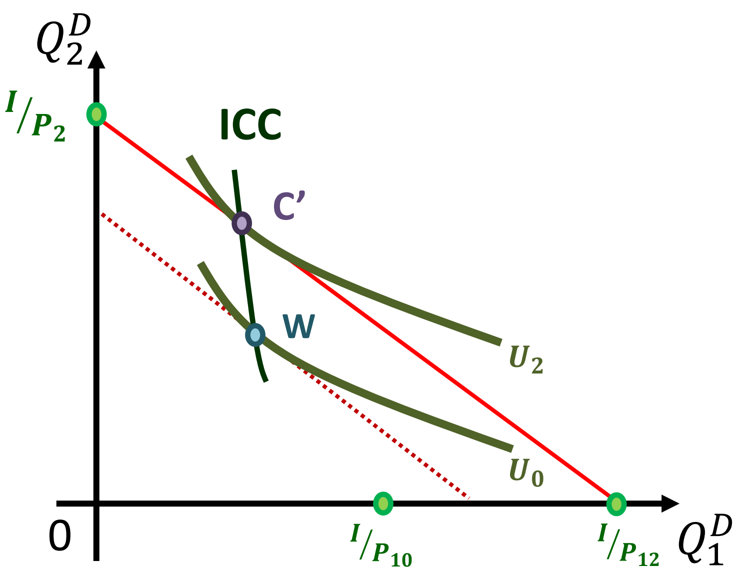
- **Case 1: Positive income effect**




- As the purchase power of the income I increases (i.e., the price of the product 1 falls), the consumer's utility can be increased by buying more product 1. As the result, the product 1 is a normal good, and it has positive income effect.

- The curve connecting W and C is income-consumption curve, ICC. If the curve has positive slope, it indicates that the product 1 has positive income effect

- **Case 2：Negative income effect**




- Although the purchase power of the income I increases, the consumer's utility is increased by buying less product 1. As the result, the product 1 is an inferior good, and it has negative income effect.

- If the product 1 is an inferior good, the slope of the ICC curve is negative.

- Whether the good is normal or inferior depends on the shape of indifference curves

# Brief Summary

- Other things being equal, the way how price changes affect a consumer's optimal decisions on quantity demanded includes two effects:

  (1) Substitution effect：

  A fall in price of one product makes other products more expensive relative to this one. A consumer will buy more of the cheaper one but fewer of others.

  A bowed inward indifference curve always leads to this negative substitution effect

- (2) Income effect:

A fall in price of one product boosts the purchasing power of a consumer’s income: he will buy more products to increase his utility

**If the product is a normal good, it has positive income effect. That is, a consumer will buy more in order to increase its utility.**

**If the product is an inferior good, it has negative income effect. That is, a consumer will buy less in order to increase its utility**

**Whether the product is normal of inferior depends on the shape of the consumer’s indifference curves even though those curves are bowed inward**

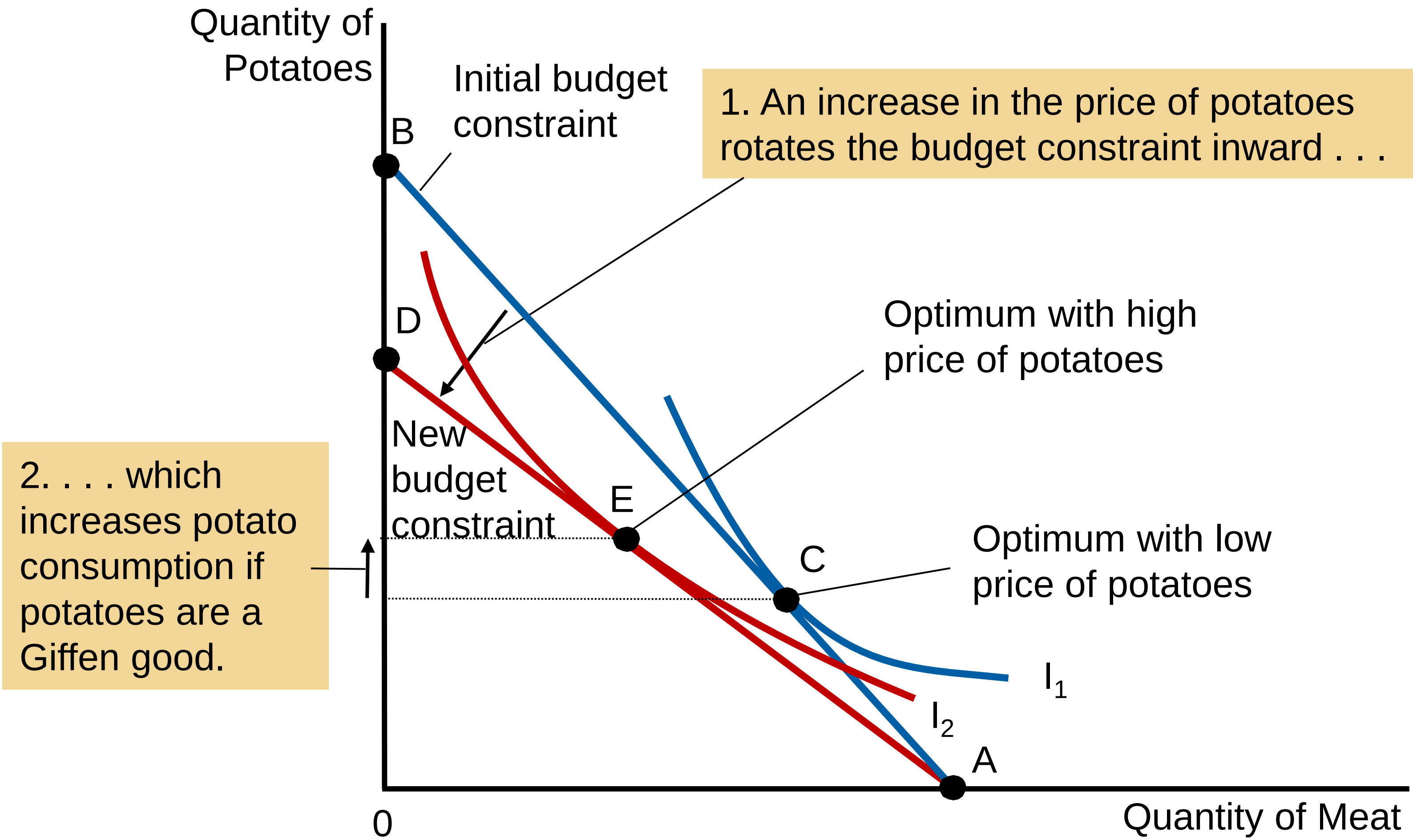
# Further Analysis: Giffen Goods

## Do all goods obey the Law of Demand?

- Suppose the goods are potatoes and meat, and potatoes are an inferior good (negative income effect).
- If price of potatoes rises,
  - Substitution effect: buy less potatoes
  - Income effect: buy more potatoes
- If rise in consumption volume implied by income effect

  > fall in consumption volume implied by substitution effect

then potatoes are a Giffen good, a good for which an increase in price raises the quantity demanded.




- In this example, when the price of potatoes rises, the consumer's optimum shifts from point C to point E. In this case, the consumer responds to a higher price of potatoes by buying less meat and more potatoes.

E_Lecture4_W2_Q1

# In-class problem

- 若所得效果為正，則需求曲線的斜率一定是負的，試證明之

**Prove that if a product has a positive income effect, then the demand curve for the product shows the negative relation between quantity demanded and price.**

- The change in optimal consumption bundle results from two effects:

  (1) Negative substitution effect：

  In order to keep utility unchanged, consumers will buy more products that become cheaper.

  (2) Income effect：

  Negative income effect: in order to rise utility, consumer will buy fewer products that become cheaper. This product is an inferior good.

  Positive income effect: in order to rise utility, consumer will buy more products that become cheaper. This product is a normal good.

- Therefore, when substitution effect is negative and income effect is positive, we obtain the negative relation between price and quantity demanded.

# 4-4: 兩種特殊的偏好與其最適消費選擇的分析

**(Two extreme cases of preferences and the corresponding analyses of optimal decisions on quantity demanded)**

（1）兩商品完全互補的偏好

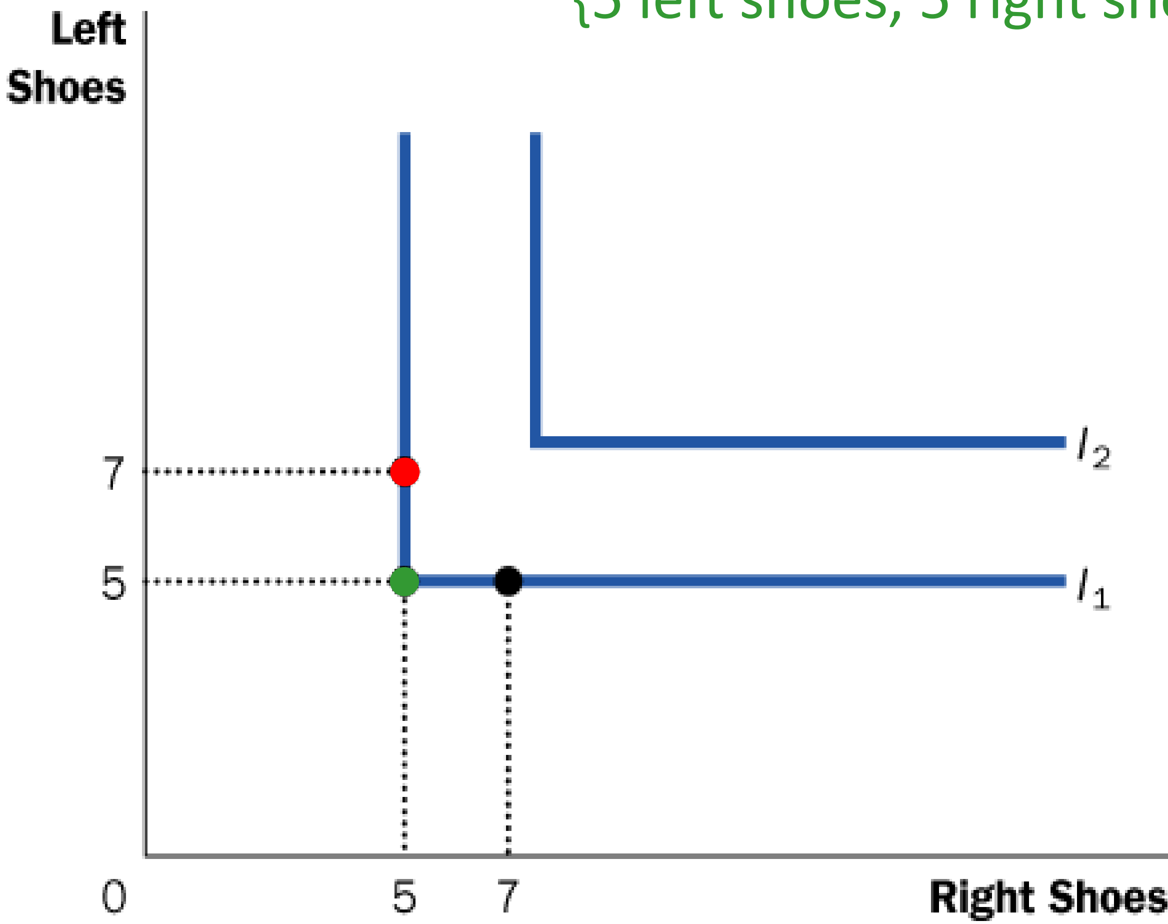
**(two goods are perfect complements)**

（2）兩商品完全替代的偏好

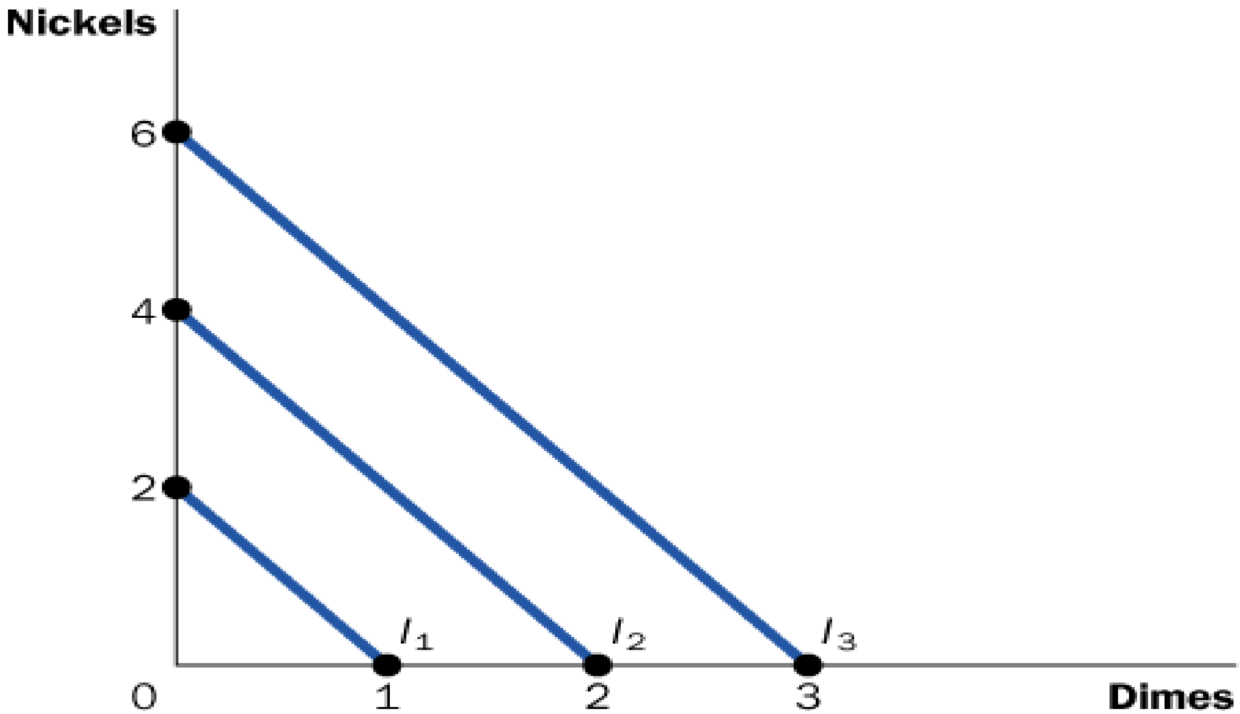
**(two goods are perfect substitutes)**

- **Perfect complements**: two goods with right-angle indifference curves

- Example: Left shoes, right shoes
  - {7 left shoes, 5 right shoes} is just as good as {5 left shoes, 5 right shoes}

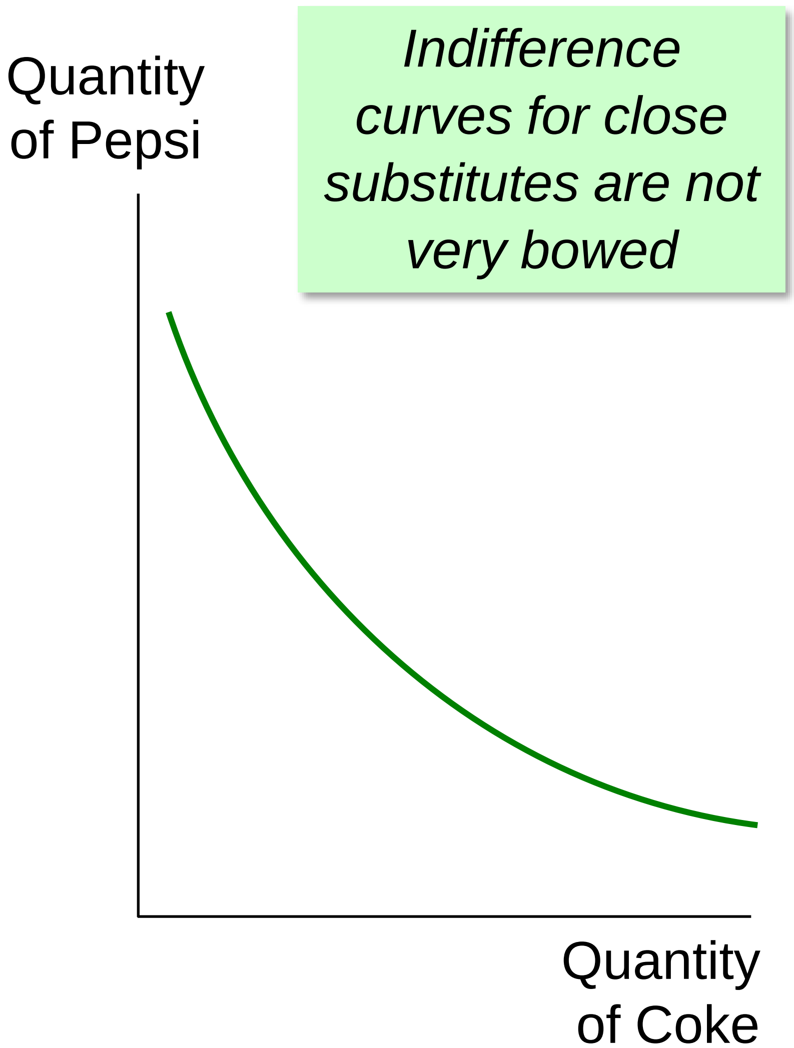


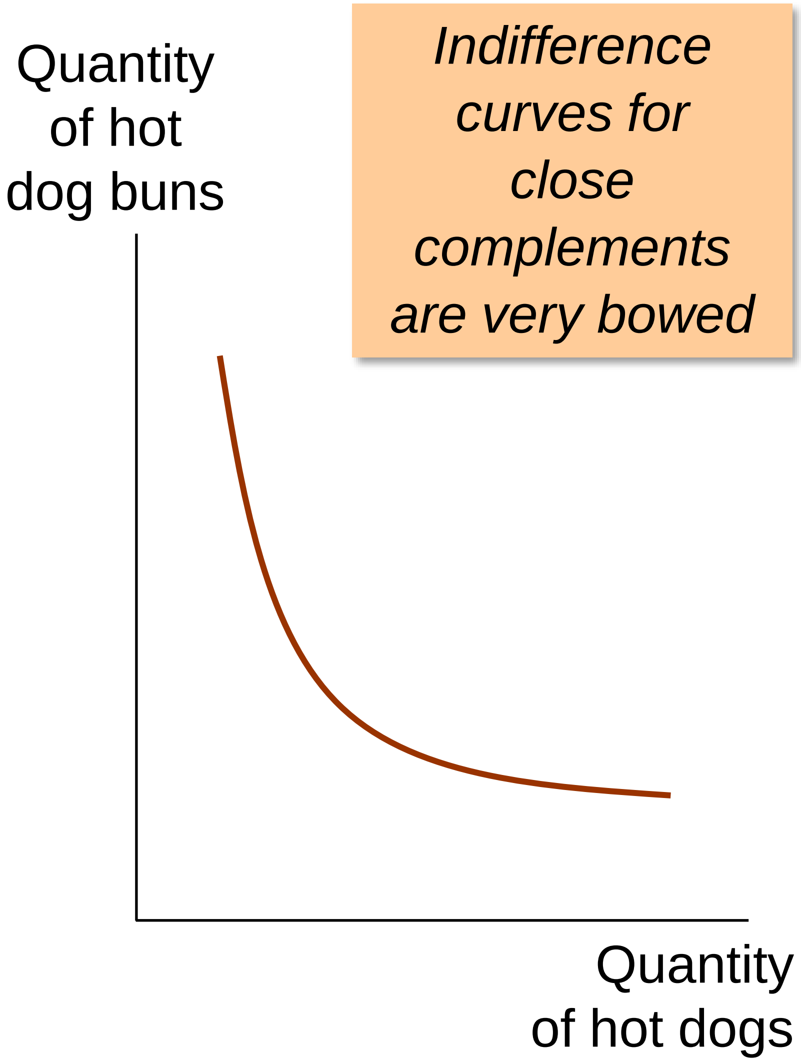
- **Perfect substitutes**: two goods with straight-line indifference curves, constant MRS
- Example: nickels and dimes
  Consumer is always willing to trade two nickels for one dime.



# Less Extreme Cases:

## Close Substitutes and Close Complements






E_Lecture4_W2_Q2-Q3